1.Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Ans. I used the ping www.youtube.com command to check my internet connection. The example average response time is 40 ms. Replace it with the average shown on your computer after running the test.

**2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.**

Traceroute to www.flipkart.com

Steps to follow:

1. Open Command Prompt (CMD) on your Windows computer.
2. Type this command and press Enter:
3. Wait for the results to appear.
4. Note the hop numbers and IP addresses shown in the output. Each hop represents a network device, usually a router, along the path to the destination.

Example table for your assignment

Your actual results may be different, so fill in the table using your own CMD output.

Hop Number	IP Address / Device
1	Your home router
2	Internet Service Provider (ISP) router
3	Next network router
4 onwards	Other routers on the route to Flipkart
Final hop	Flipkart server, if reached

3.Use the nslookup tool to find the IP address of www.zomato.com and record the result.

- Command used: `nslookup www.zomato.com`
- IP address: The address returned by your computer may vary.
- Conclusion: The `nslookup` command finds the IP address linked to a website's domain name.

4.Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Ping Test on a Local Network

Steps:

Connect your computer and your friend's laptop or phone to the same Wi-Fi.

Find the other device's IP address.

Open Command Prompt on your computer.

Type ping followed by the device's IP address. For example:

ping 192.168.1.5

Press Enter and check the result.

If the device is online: You may receive replies showing the response time in milliseconds (ms).

If the device is offline: You may see “Request timed out” because the device cannot reply. However, this can also happen if its firewall blocks ping requests.

**5.Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.**

Network Troubleshooting Tools

1) **Website not loading: I use ping www.youtube.com to check whether the website responds to my computer.**
2) **Slow connection: I use tracert www.flipkart.com to check the route and find where delays may happen.**
3) **DNS issues: I use nslookup www.zomato.com to find the website's IP address.**
4) **In my ping test example, the average response time was 40 ms, but my actual result may be different.**
5) **The traceroute command shows the network hops between my computer and Flipkart.**
6) **The nslookup command helps check whether Zomato's domain name can be resolved to an IP address.**
7) **Conclusion: Ping checks connectivity, traceroute checks the network path, and nslookup checks DNS resolution.**